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By

B.Dhanush kumar

DECLARATION

We hereby declare that the project work titled **"Integrated Turbocharger-Assisted Exhaust Energy Recovery System for Electrical Power Generation in a V6 Engine"** is a bonafide work carried out by us in partial fulfillment of the requirements for this internship.

This work has been carried out under the guidance of **Prof. Radhakumari Ma'am** and **Mr. Mairala Chinnarao**. We further declare that this report has not been submitted previously to any other organization.

B.Dhanush kumar

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to **Sure Trust Organization** for providing us with the opportunity and platform to pursue this internship and carry out our project work on “Origami-Based Collapsible Roof System.” The exposure and learning experience gained through this initiative have been invaluable in enhancing our technical knowledge and practical skills.

We are deeply thankful to our faculty guide, **Prof. Radhakumari Ma’am**, for her constant encouragement, valuable suggestions, and continuous support throughout the course of this project. Her guidance has played a crucial role in shaping our understanding and helping us successfully complete this work.

We extend our heartfelt thanks to our mentor, **Mr. Mairala Chinnarao**, for his expert guidance, practical insights, and consistent supervision at every stage of the project. His clarity in explaining concepts and willingness to assist us greatly contributed to the successful execution of this work.

We also express our sincere appreciation to the Head of the Department and all the faculty members of the Civil Engineering Department for their support and encouragement.

Finally, we would like to thank all our batchmates and team members for their cooperation, teamwork, and support, which helped us complete this project successfully.

TABLE OF CONTENTS

ABSTRACT

Internal combustion engines lose a significant amount of energy through exhaust gases. This project proposes the Design and Development of an Integrated Turbocharger-Assisted Exhaust Energy Recovery System with Electrical Power Generation for a V6 Engine.

The proposed system utilizes exhaust-gas energy to rotate a turbine, which drives a compressor and a motor-generator through a common shaft. The compressor supplies compressed air to the engine intake, while the motor-generator converts a portion of the recovered mechanical energy into electrical power. The complete system is designed using SolidWorks, including the V6 engine, exhaust manifold, turbine, compressor, shaft, generator, and supporting frame.

Material selection, engineering calculations, and basic design analysis are performed to evaluate the feasibility of the system. The proposed concept aims to improve exhaust energy utilization, overall engine efficiency, and electrical power generation while providing practical experience in mechanical design and CAD modelling.

CHAPTER 1: INTRODUCTION

Internal combustion engines are widely used in automobiles and other mechanical applications, but a considerable portion of the fuel energy is lost through exhaust gases. These exhaust gases contain useful thermal and kinetic energy that can potentially be recovered. A conventional turbocharger utilizes a portion of this exhaust energy to drive a turbine connected to a compressor, thereby increasing the intake-air pressure and improving engine performance.

This project proposes an Integrated Turbocharger-Assisted Exhaust Energy Recovery System with Electrical Power Generation for a V6 Engine. In the proposed concept, exhaust gases rotate a turbine connected through a common shaft to the compressor and motor-generator. The compressor supplies compressed air to the engine, while the motor-generator converts a portion of the recovered mechanical energy into electrical power.

The complete system is designed using SolidWorks, including the V6 engine, exhaust manifold, turbine, compressor, shaft, generator, and supporting frame. Material selection, engineering calculations, and design analysis are also considered to evaluate the feasibility of the proposed system. The project aims to improve waste-energy utilization and overall engine efficiency while developing practical skills in mechanical design and CAD modelling.

CHAPTER 2: OBJECTIVES

The main objective of this project is to **design and develop an integrated turbocharger-assisted exhaust energy recovery system with electrical power generation for a V6 engine**. The project aims to utilize the otherwise wasted energy present in engine exhaust gases by directing them through a turbine.

The turbine converts the exhaust-gas energy into rotational mechanical energy and transfers it through a common shaft to the compressor and motor-generator. The compressor is intended to improve the intake-air supply to the engine, while the motor-generator converts a portion of the recovered mechanical energy into electrical power. The complete system will be designed and assembled using SolidWorks, including the V6 engine, exhaust manifold, turbine, compressor, shaft, generator, and supporting frame.

The project also aims to select suitable materials, perform engineering calculations, evaluate stresses and deformation of critical components, and study the overall feasibility of the proposed system. Through this project, the objective is to improve waste-energy utilization, mechanical design, and electrical power recovery while gaining practical experience in CAD modelling, engineering analysis, and automotive system development.

CHAPTER 3: LITERATURE REVIEW

Several researchers have studied the recovery of waste energy from internal combustion engine exhaust gases to improve overall engine efficiency. **Aghaali and Ångström** reviewed turbo-compounding technology and reported its potential for recovering exhaust energy from internal combustion engines.

Feneley, Pesiridis, and Andwari investigated variable-geometry turbocharger technologies for exhaust energy recovery and engine boosting.

Zhuge et al. studied electric turbo-compounding systems for gasoline engines, focusing on recovering exhaust energy for electrical power generation.

Thompson et al. investigated turbogenerating systems and demonstrated the technical potential of converting exhaust-gas energy into useful electrical power.

Wei et al. compared electric turbo-compounding systems and evaluated their potential for improving engine energy utilization. Based on these studies, turbocharging and exhaust energy recovery are established engineering approaches, while integrating the **turbine, compressor, and motor-generator** into a single system provides an opportunity for further design investigation. Therefore, this project focuses on developing a **SolidWorks-based conceptual design for a V6 engine**, supported by material selection, engineering calculations, and simulation.

CHAPTER 4: PROBLEM STATEMENT

Internal combustion engines experience considerable energy losses through exhaust gases, resulting in inefficient utilization of the fuel energy supplied to the engine. Conventional turbochargers recover a portion of the exhaust energy to drive a compressor, but additional recoverable energy may remain unused. At the same time, conventional engine-driven electrical generation places an additional mechanical load on the engine. Therefore, there is a need to investigate an integrated system that can utilize exhaust-gas energy more effectively while also producing useful electrical power. This project proposes the design of an integrated turbocharger-assisted exhaust energy recovery system with electrical power generation for a V6 engine, in which an exhaust-driven turbine is mechanically connected to a compressor and motor-generator through a suitable shaft arrangement.

CHAPTER 5: METHODOLOGY

The methodology of this project involves the design and development of an integrated turbocharger-assisted exhaust energy recovery system with electrical power generation for a V6 engine.

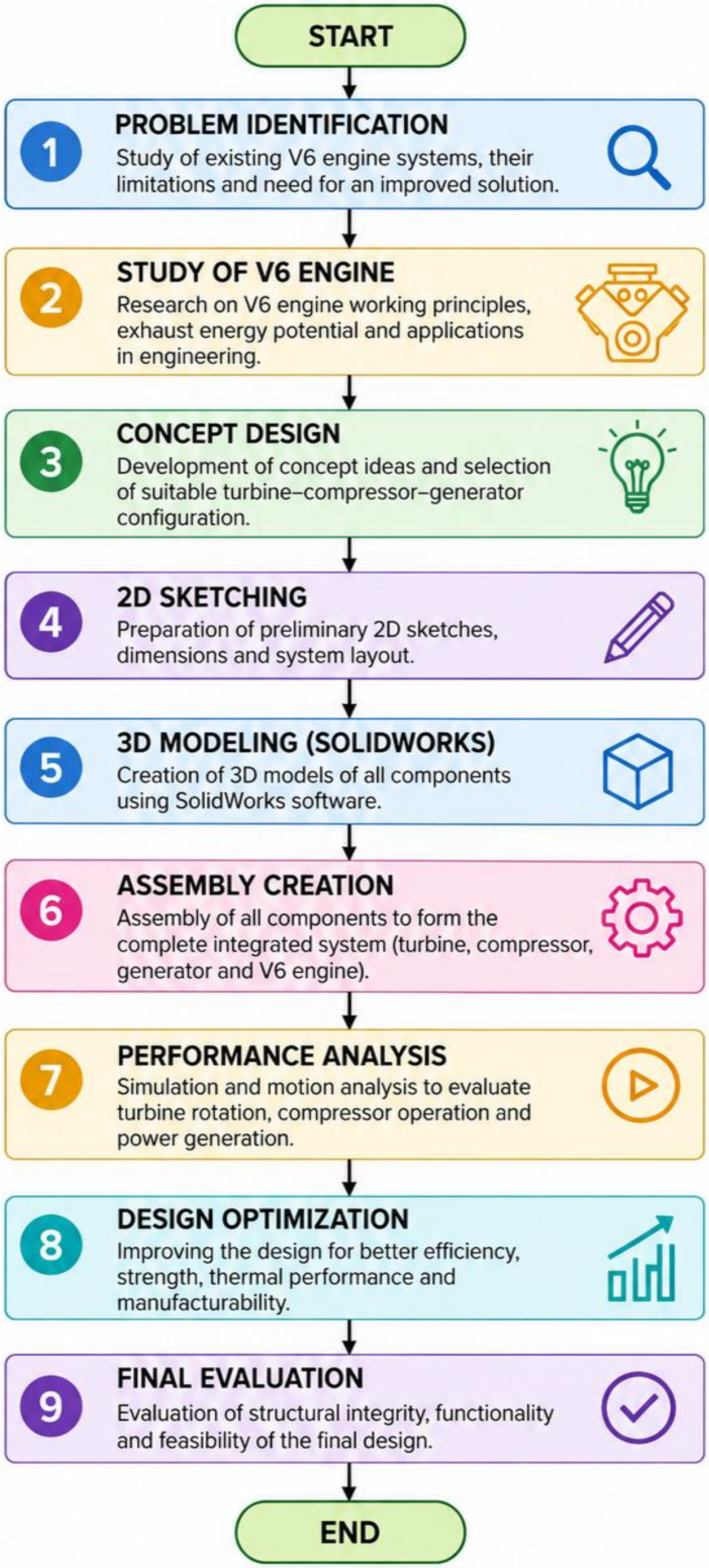
Initially, a literature study is carried out to understand existing turbocharger, turbo-compounding, and exhaust energy recovery technologies. A suitable V6 engine is then selected and its operating parameters are considered for the system design. The proposed system consists of an exhaust manifold, turbine, common shaft, compressor, and motor-generator.

The exhaust gases rotate the turbine, and the recovered rotational energy is transferred through the shaft to drive the compressor and generator. The major components are designed and modelled individually using SolidWorks, followed by complete assembly of the system.

Suitable materials are selected based on strength, temperature resistance, weight, and durability. Engineering calculations are performed to determine exhaust energy, turbine power, compressor requirements, shaft torque, and expected electrical power generation. Finally, the assembled model is evaluated using SolidWorks Simulation where applicable to study stress and deformation of critical components, and the overall feasibility of the proposed energy recovery system is assessed.

METHODOLOGY FLOWCHART

INTEGRATED TURBOCHARGER-ASSISTED EXHAUST ENERGY – ELECTRICAL POWER GENERATION (V6)



CHAPTER 6: SYSTEM OVERVIEW

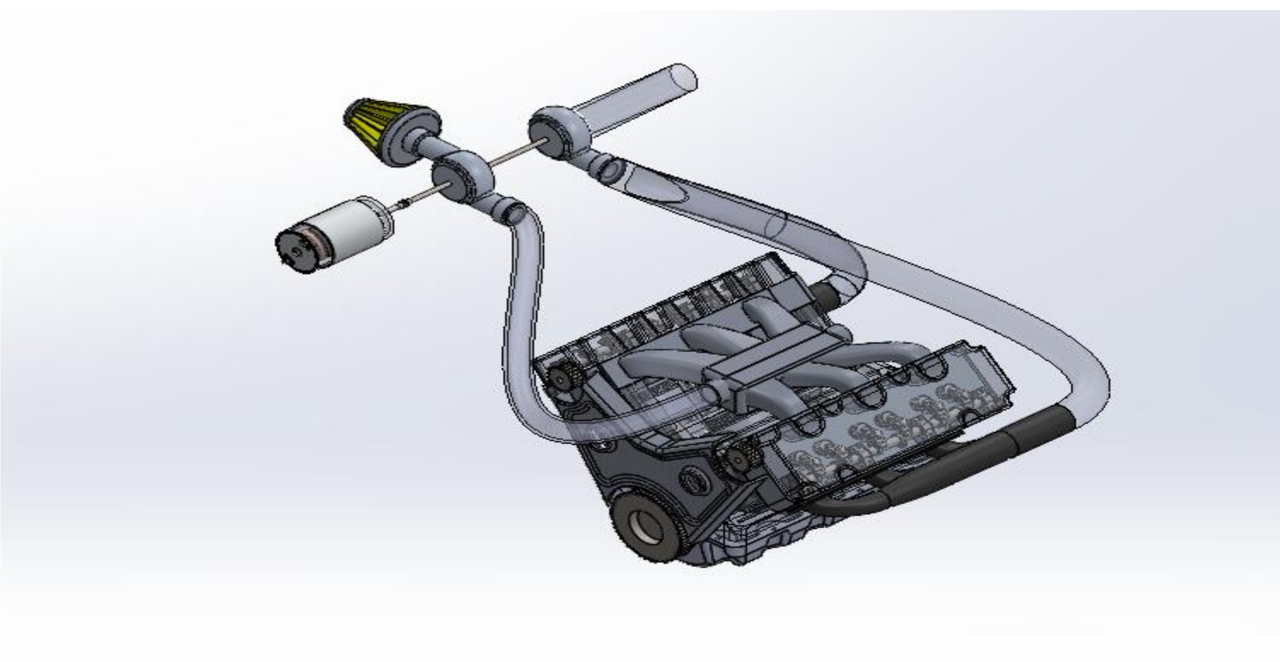
The proposed system consists of a **V6 engine** that acts as the primary power source. The engine produces high-temperature and high-velocity exhaust gases, which are directed toward the turbine through the exhaust manifold.

The **exhaust manifold and turbine** form the exhaust energy recovery section. The manifold collects exhaust gases from the V6 cylinders and directs them to the turbine. The turbine converts the energy of the exhaust gases into rotational mechanical energy.

The **common shaft and compressor** are connected to the turbine. The shaft transfers the rotational motion from the turbine to the compressor.

A **motor-generator** is connected to the rotating shaft to utilize a portion of the recovered mechanical energy for electrical power generation. The generated electrical power can be supplied to suitable electrical loads or stored in a battery through a suitable control system.

The **supporting frame and mounting structure** hold the turbine, compressor, shaft, and motor-generator in their required positions. The complete system is designed and assembled in **SolidWorks** to study the integration, dimensions, motion, and structural feasibility of the proposed system.



6.1 COMPONENT DESIGN

The V6 engine is designed as the main power source of the system. It consists of two cylinder banks arranged in a V configuration, with the required engine components modelled to provide a suitable platform for integrating the exhaust energy recovery system.

The exhaust manifold is connected to the engine exhaust ports and is designed to collect and guide the exhaust gases toward the turbine. The curved manifold pipes provide a continuous flow path from the engine to the turbine.

The turbine is positioned in the exhaust flow path. It receives high-energy exhaust gases from the manifold and converts the exhaust energy into rotational motion. The turbine is connected to the rotating shaft of the proposed system.

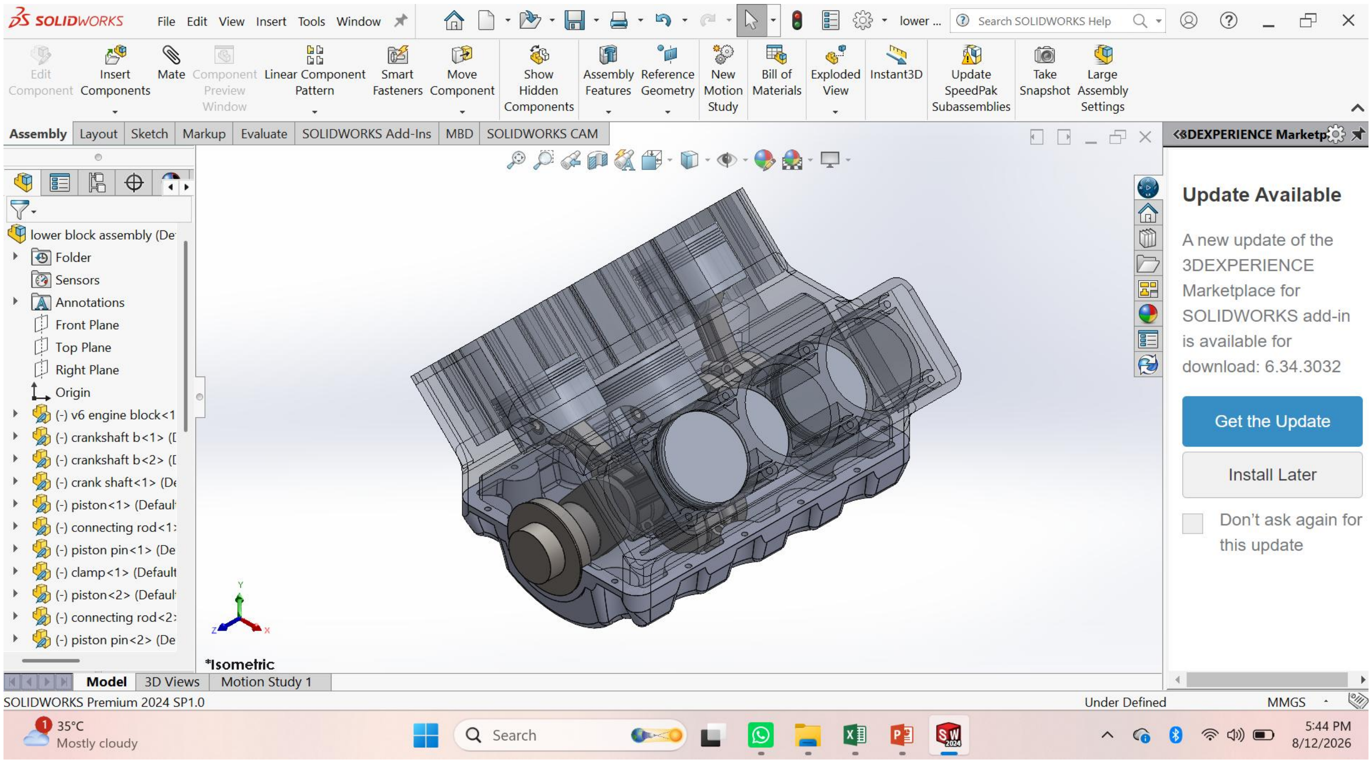
The compressor is connected to the turbine through the shaft and utilizes the recovered rotational energy to compress the incoming air. The compressed air is then directed toward the engine intake through the intake pipe.

The shaft and coupling mechanism transmit rotational motion from the turbine to the compressor and the motor-generator. The shaft is designed to maintain proper alignment and transmit the required torque during operation.

The motor-generator is connected to the rotating shaft through a coupling. During energy recovery, it operates as a generator and converts a portion of the mechanical energy into electrical power.

The supporting frame and mounting components provide structural support for the turbine, compressor, shaft, and generator. They also help maintain proper alignment and stability of the complete system.

The complete system is modelled and assembled in SolidWorks, as shown in the figure, to study the component arrangement, connections, dimensions, and overall feasibility of the proposed V6 exhaust energy recovery system.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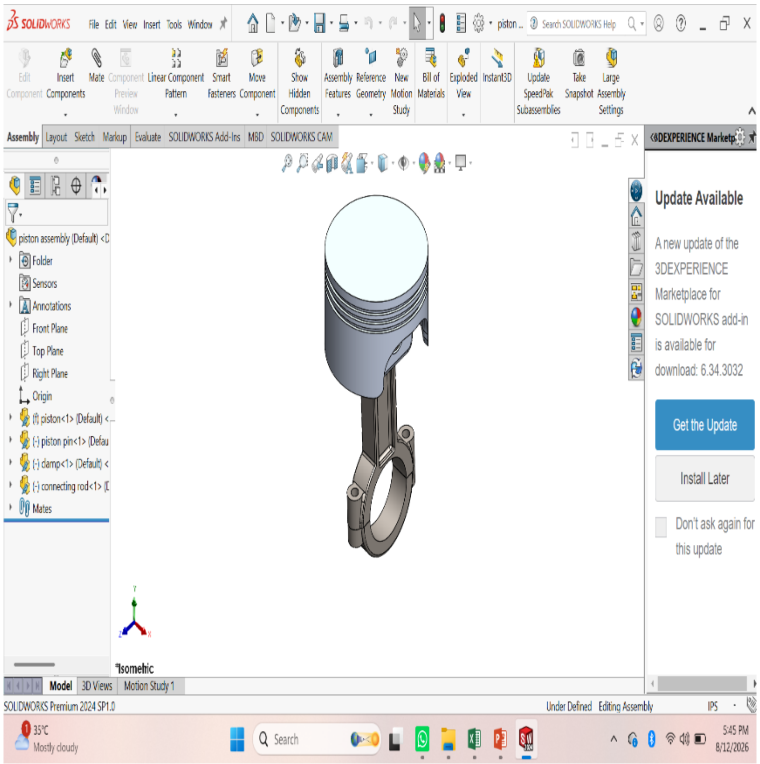
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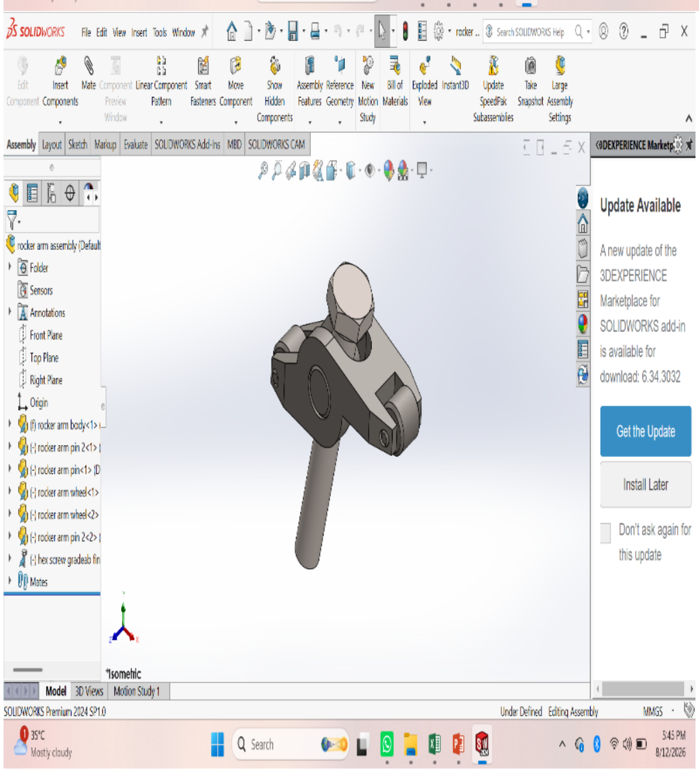
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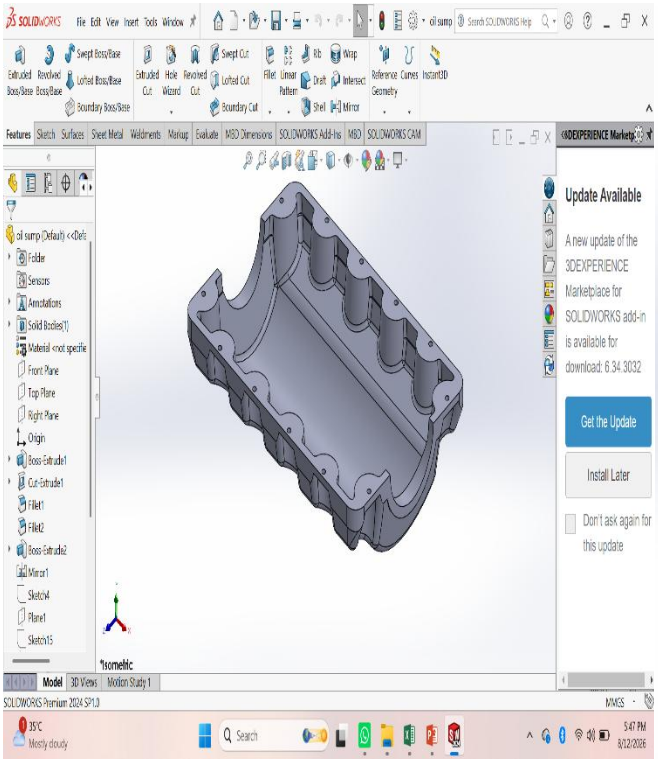

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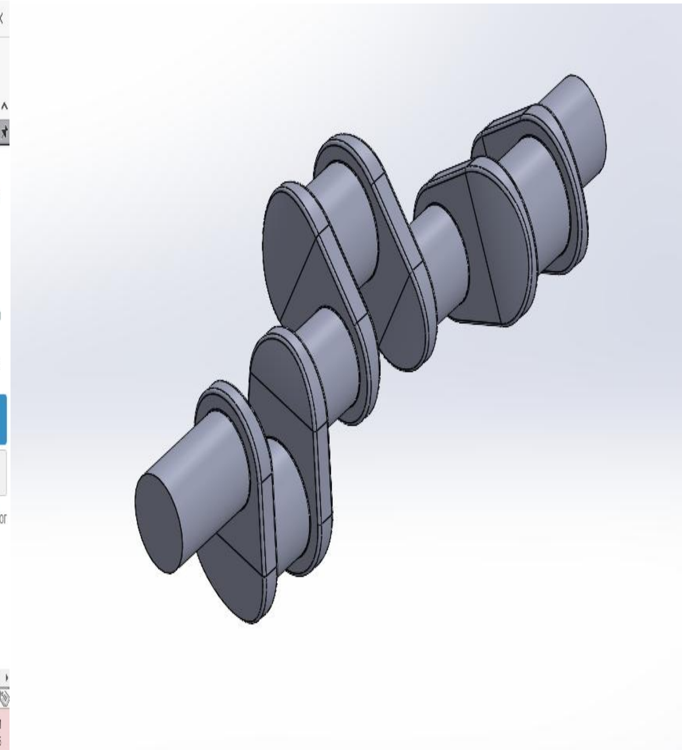
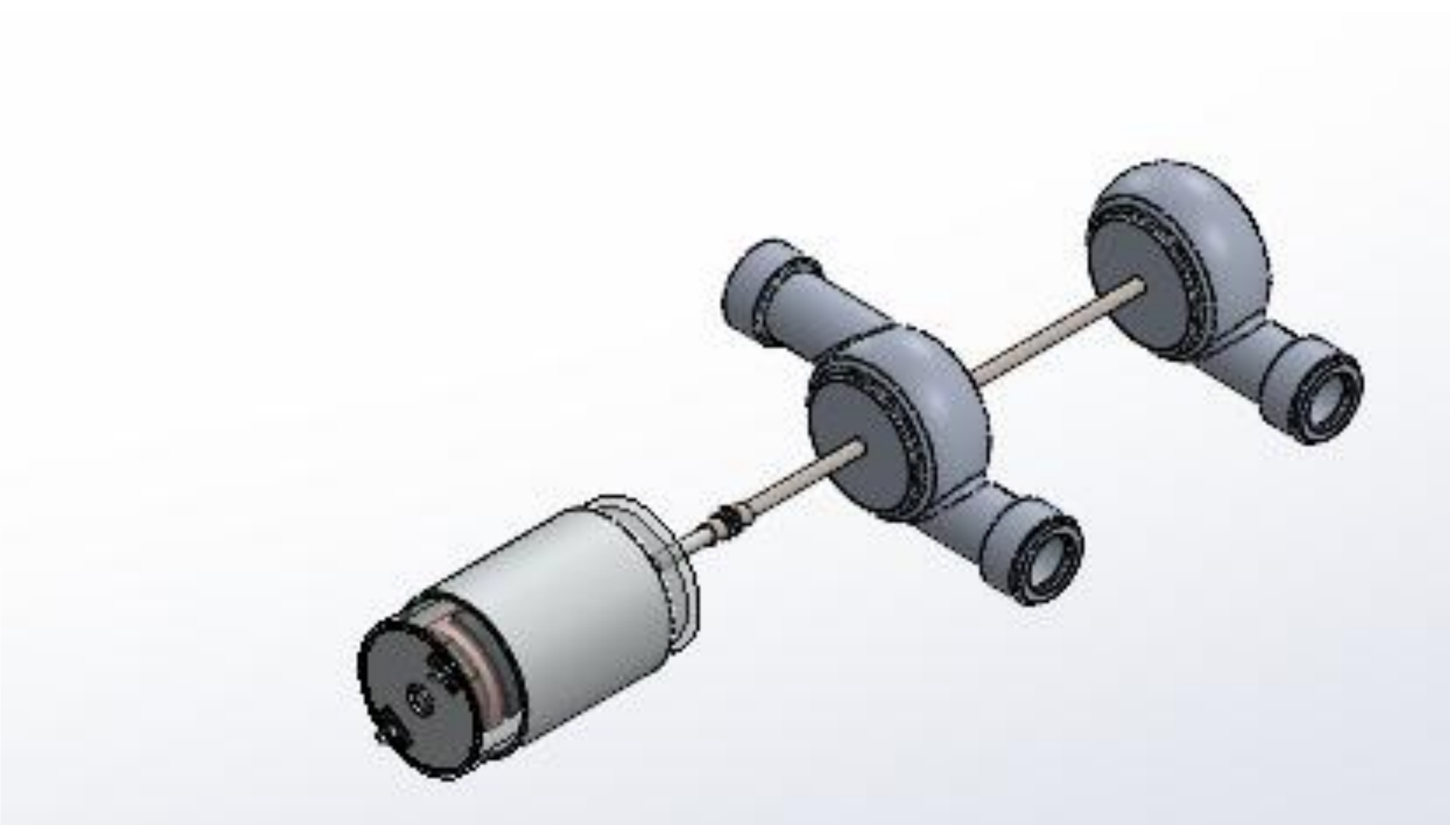




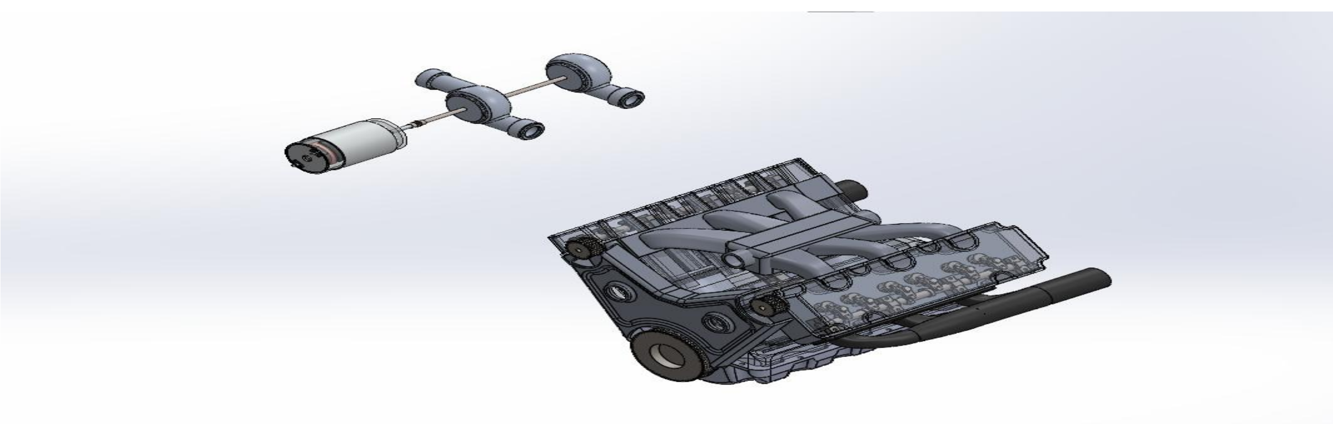
6.2 DESIGN AND MODELING

The design and modelling process was carried out using **SolidWorks software**. Each component of the V6 exhaust energy recovery system was modelled individually with suitable dimensions to ensure proper integration and functionality.

The V6 engine and exhaust manifold were designed first. The manifold was modelled with curved pipes to collect exhaust gases from the engine cylinders and guide them toward the turbine.The turbine and compressor were designed based on the required flow path and mounting arrangement. The turbine was positioned at the exhaust outlet, while the compressor was connected to the turbine through a common shaft to utilize the recovered rotational energy.

The shaft and motor-generator connection were designed to transmit rotational motion from the turbine to the compressor and generator. Suitable coupling and bearing arrangements were incorporated to maintain alignment and support high-speed rotation.

The supporting frame was modelled to hold the complete turbocharger and generator assembly securely. Finally, all components were combined into a complete SolidWorks assembly, and the assembly was checked for proper alignment, interference, and overall system integration. Motion and basic simulation studies can be performed to evaluate the rotational movement and structural feasibility of the proposed design.



CHAPTER 7: WORKING PRINCIPLE

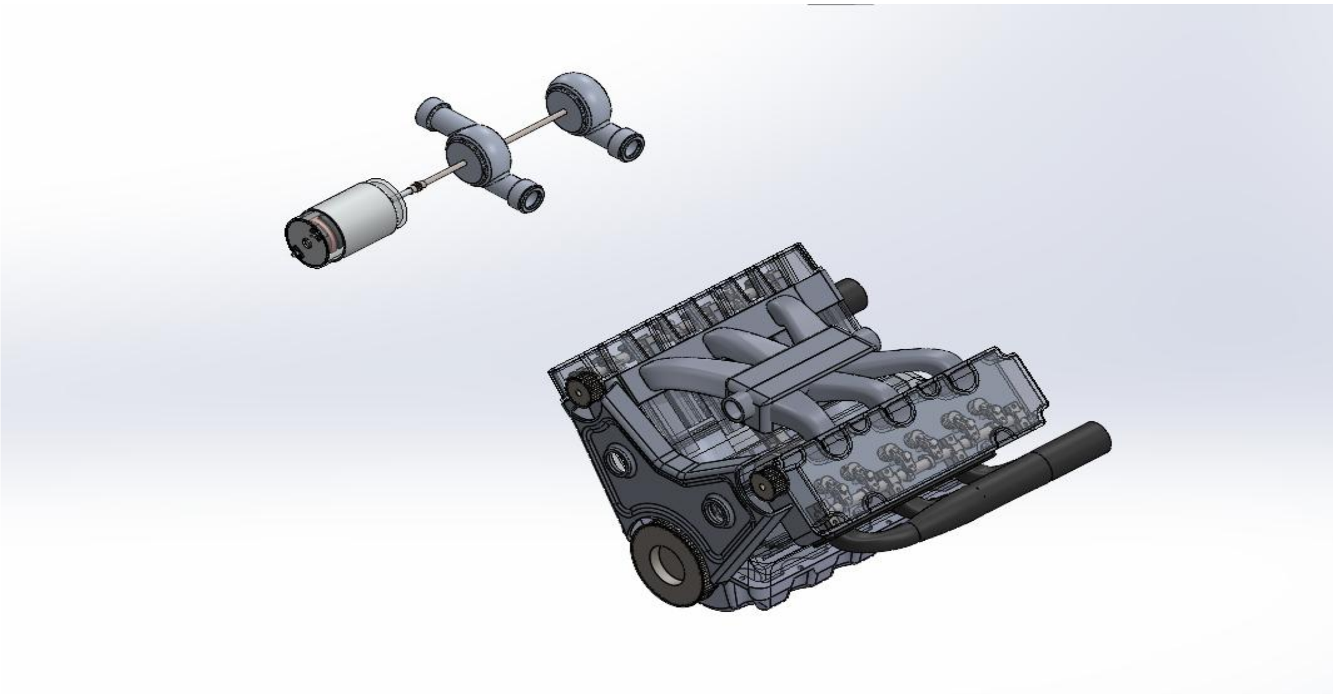
The proposed system operates on the principle of **exhaust energy recovery and turbocharging**. During the combustion process in the V6 engine, high-pressure and high-temperature exhaust gases are produced. These gases are collected from the cylinder banks through the **exhaust manifold** and directed toward the turbine.

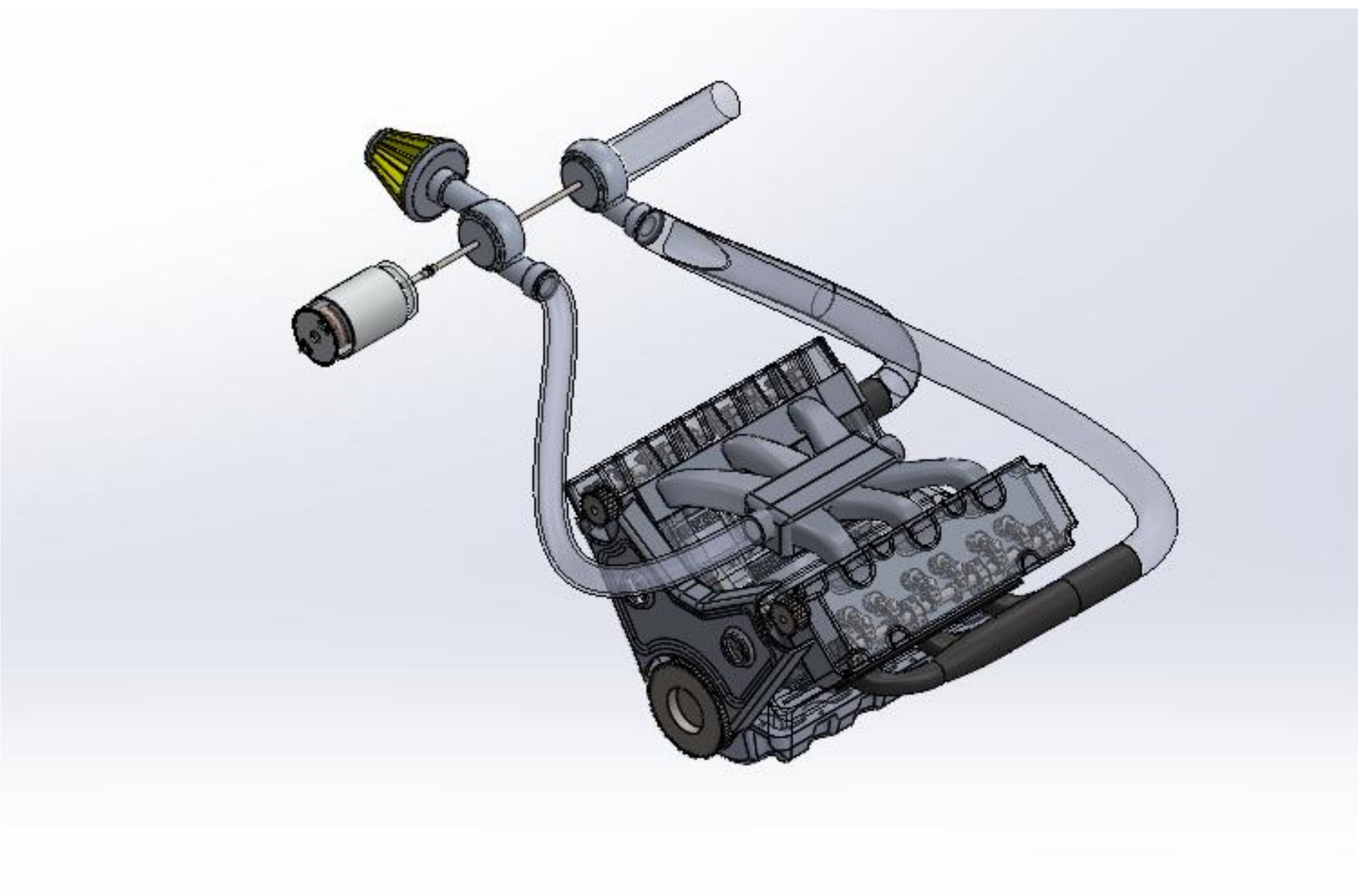
As the exhaust gases pass through the turbine, their pressure and kinetic energy are converted into rotational mechanical energy, causing the turbine wheel to rotate at high speed. The turbine is connected to a **common shaft**, which transfers this rotational motion to the compressor and motor-generator.

The **compressor** uses the rotational energy received from the turbine to draw in fresh atmospheric air, compress it, and deliver the pressurized air toward the engine intake. The increased intake-air supply can support better cylinder filling and combustion, subject to proper boost and engine-control limits.

At the same time, the **motor-generator** connected to the shaft extracts a controlled portion of the available shaft power and converts it into electrical energy. The generated electricity can be conditioned through a suitable controller and supplied to electrical loads or stored in a battery. The amount of power extracted by the generator must be carefully controlled because excessive generator load can reduce turbine speed, decrease compressor performance, and increase exhaust back pressure.

Therefore, the proposed system requires proper matching of the **turbine, compressor, shaft, and generator** to achieve effective energy recovery. The complete working arrangement is designed in SolidWorks and evaluated through engineering calculations and analysis to determine its feasibility and performance.





CHAPTER 8:

8.1 APPLICATIONS

The proposed integrated turbocharger-assisted exhaust energy recovery system can be applied in automotive and heavy-duty vehicles such as cars, trucks, buses, and performance vehicles equipped with V6 or larger engines.

The system can utilize exhaust energy for turbocharging while generating additional electrical power, which can be used for vehicle electrical loads or battery charging.

It can also be explored for hybrid vehicles, diesel engines, stationary power-generation engines, and industrial engines, where continuous operation and high exhaust-energy availability provide greater opportunities for energy recovery. The concept has potential to contribute to improved energy utilization, reduced energy losses, and development of more efficient engine systems.

8.2 ADVANTAGES

The proposed integrated turbocharger-assisted exhaust energy recovery system provides several advantages. It utilizes waste exhaust energy that would otherwise be released into the atmosphere and converts a portion of it into useful mechanical and electrical energy.

The compressor can improve the intake-air supply to the V6 engine, while the motor-generator can produce additional electrical power.

The system can help improve overall energy utilization, reduce the mechanical load associated with conventional engine-driven electrical generation when appropriately integrated, and make better use of the engine's available exhaust energy. It also provides a compact integrated design combining turbocharging and electrical power generation.

8.3 LIMITATIONS

The proposed has certain limitations. The amount of electrical power generated depends strongly on engine speed, load, exhaust-gas temperature, and exhaust mass flow rate. At low engine speeds and low loads, the available exhaust energy may be insufficient for significant power generation.

Another major limitation is exhaust back pressure**.** Installing additional turbine or recovery components in the exhaust system can restrict gas flow and increase pumping losses, which may reduce engine efficiency and output power if the system is not properly matched.

Finally, the proposed system may increase design complexity, cost, weight, and maintenance requirements compared with a conventional exhaust system. Further experimental testing under different V6 engine speeds and loads is required to determine the actual net power generation and overall efficiency of the system.

8.4 FUTURE SCOPE

The future scope of the system can be further improved by using a variable-geometry turbine and a high-efficiency generator to increase energy recovery under different engine operating conditions. CFD analysis and experimental testing can be used to optimize the turbine, compressor, exhaust manifold, and generator arrangement. Advanced materials can also be used to withstand high exhaust temperatures and rotational speeds while reducing the overall weight of the system. An automatic electronic control system can be developed to regulate turbine speed and electrical output efficiently. The generated electrical energy can be stored in a battery and used for auxiliary vehicle systems. Further research can also combine this system with other waste-heat recovery technologies to improve overall engine efficiency and reduce fuel consumption and emissions.

CHAPTER 14: CONCLUSION

The project successfully demonstrates the design and development of an Integrated Turbocharger-Assisted Exhaust Energy Recovery System for Electrical Power Generation in a V6 Engine. The system utilizes the energy available in exhaust gases to drive a turbine connected to a generator, enabling useful electrical power generation from otherwise wasted exhaust energy.

The proposed system provides an efficient approach to waste-energy recovery while maintaining the primary function of the V6 engine. The design offers potential for improving overall energy utilization, reducing energy losses, and supporting the electrical requirements of automotive systems.

The developed concept has significant potential for further optimization through improved turbine and generator design, advanced control systems, and experimental testing. It represents a promising step toward efficient waste-energy recovery and sustainable power generation in modern automotive engines.